

HT Matrices 2

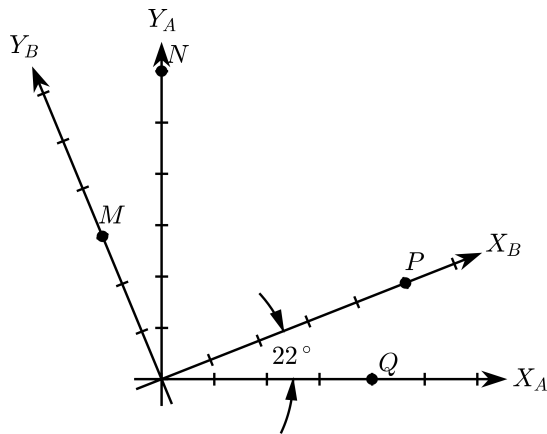
EGR 445/545

GVSU

Three Main Things for Today

- go deeper on HT matrices so you are truly ready for LA 5
- go over two link inverse kinematics so you are ready for project 1
- project time
 - two byte serial
 - controlling servos from Python

Review: LA 3



What do you expect for the signs of the following vectors?:

- Q_B
- P_A
- N_B
- M_A

The basis of HT matrices: Eqns. 2.17 - 19

$$\text{Eqn. 2.17: } {}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

$$\text{Eqn. 2.18: } \begin{bmatrix} {}^A \mathbf{P} \\ 1 \end{bmatrix} = {}^A_B \mathbf{T} \begin{bmatrix} {}^B \mathbf{P} \\ 1 \end{bmatrix}$$

$$\text{Eqn. 2.19: } \begin{bmatrix} {}^A \mathbf{P} \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B \mathbf{R} & {}^A \mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B \mathbf{P} \\ 1 \end{bmatrix}$$

Goal

- Our goal is to model any robot using HT matrices like so:

$${}^0P_{tip} = {}^0_1\mathbf{T} \cdot {}^1_2\mathbf{T} \cdot {}^2_3\mathbf{T} \cdot {}^3_4\mathbf{T} \cdot {}^4_5\mathbf{T} \cdot {}^5_6\mathbf{T} \cdot {}^6P_{tip}$$

- the HT matrices will come from eqn. 3.6 and the Denavit-Hartenberg convention
 - we are using JJ Craig's Introduction to Robotics book

Frame attachment and mapping

the big idea:

- we attach frames by starting at the base of the robot and working our way out to the tip
- we know the tip in the last frame and need to map it back to frame 0
- possibly confusing result: frame attachment and mathematical mapping go in opposite directions

Need

- we need to combined rotation and translation into one matrix multiplication operation:

$${}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

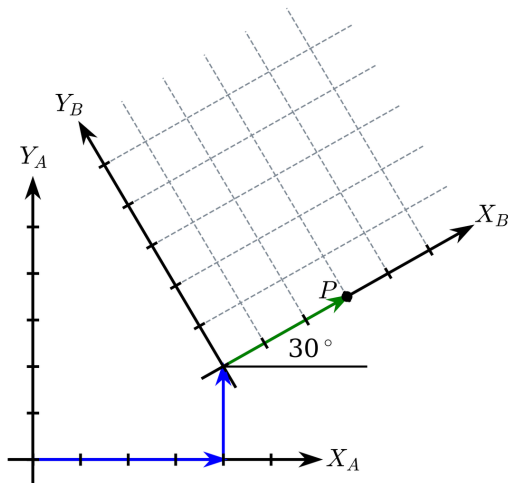
Claim

- a 4x4 HT matrix of this form meets our need for combining rotation and translation:

$${}^A_B\mathbf{T} = \begin{bmatrix} {}^A_B\mathbf{R} & {}^A\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Adding Vectors

- we cannot add vectors in different frames:



Relationship between frames A and B

Translate First, then Rotate:

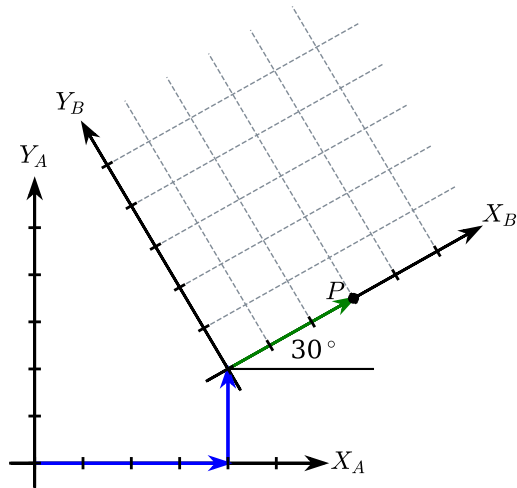
- start with frames A and B coincident
- translate from B by ${}^A P_{BORG}$
- rotate frame B
- note that the mapping is from B back to A

Mathematical Order

- note the order of operations:

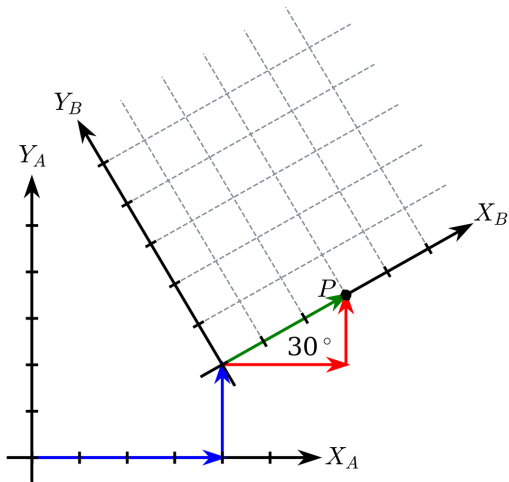
$${}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

- rotate into frame A
- add the vectors in frame A components



Adding in the Same Frame

- once the vector components are in the same frame, we can add them legally:



HT Matrix Math

$${}^A P = {}^A_B \mathbf{R} \cdot {}^B P + {}^A P_{BORG}$$

becomes

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B \mathbf{R} & {}^A \mathbf{P}_{BORG} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

HT Math R_z Example

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & x \\ \sin(\theta) & \cos(\theta) & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x \cos(\theta) - {}^B P_y \sin(\theta) + x \\ {}^B P_x \sin(\theta) + {}^B P_y \cos(\theta) + y \\ {}^B P_z + z \\ 1 \end{bmatrix}$$

Why is the bottom row necessary?

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 & x \\ \sin(\theta) & \cos(\theta) & 0 & y \\ 0 & 0 & 1 & z \end{bmatrix} \cdot \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x \cos(\theta) - {}^B P_y \sin(\theta) + x \\ {}^B P_x \sin(\theta) + {}^B P_y \cos(\theta) + y \\ {}^B P_z + z \end{bmatrix}$$

- we have the correct answer for ${}^A P$, but the dimensions of the answer are 3x1
 - we could not legally multiply by the next 4x4 if we needed to do a repeated mapping:

$${}^0 P_{tip} = {}^0_1 \mathbf{T} \cdot {}^1_2 \mathbf{T} \cdot {}^2_3 \mathbf{T} \cdot {}^3_4 \mathbf{T} \cdot {}^4_5 \mathbf{T} \cdot {}^5_6 \mathbf{T} \cdot {}^6 P_{tip}$$

A General HT Matrix

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x r_{11} + {}^B P_y r_{12} + {}^B P_z r_{13} + x \\ {}^B P_x r_{21} + {}^B P_y r_{22} + {}^B P_z r_{23} + y \\ {}^B P_x r_{31} + {}^B P_y r_{32} + {}^B P_z r_{33} + z \\ 1 \end{bmatrix}$$

What if the bottom row were different?

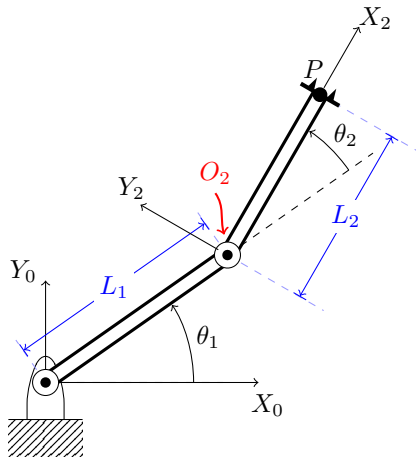
$$\begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ u & v & w & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^B P_x \\ {}^B P_y \\ {}^B P_z \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P_x r_{11} + {}^B P_y r_{12} + {}^B P_z r_{13} + x \\ {}^B P_x r_{21} + {}^B P_y r_{22} + {}^B P_z r_{23} + y \\ {}^B P_x r_{31} + {}^B P_y r_{32} + {}^B P_z r_{33} + z \\ {}^B P_x u + {}^B P_y v + {}^B P_z w + 1 \end{bmatrix}$$

- the bottom row must be $[0, 0, 0, 1]$ for subsequent multiplications

HT Matrices and my robotics module

- `robotics.HTz`
 - `robotics.HTx`
 - `robotics.HTy`
- `robotics.HT_from_R_and_Porg`
 - this is foundational
 - what does this code do?

Two-Link Example



Review: Find ${}^A P$

